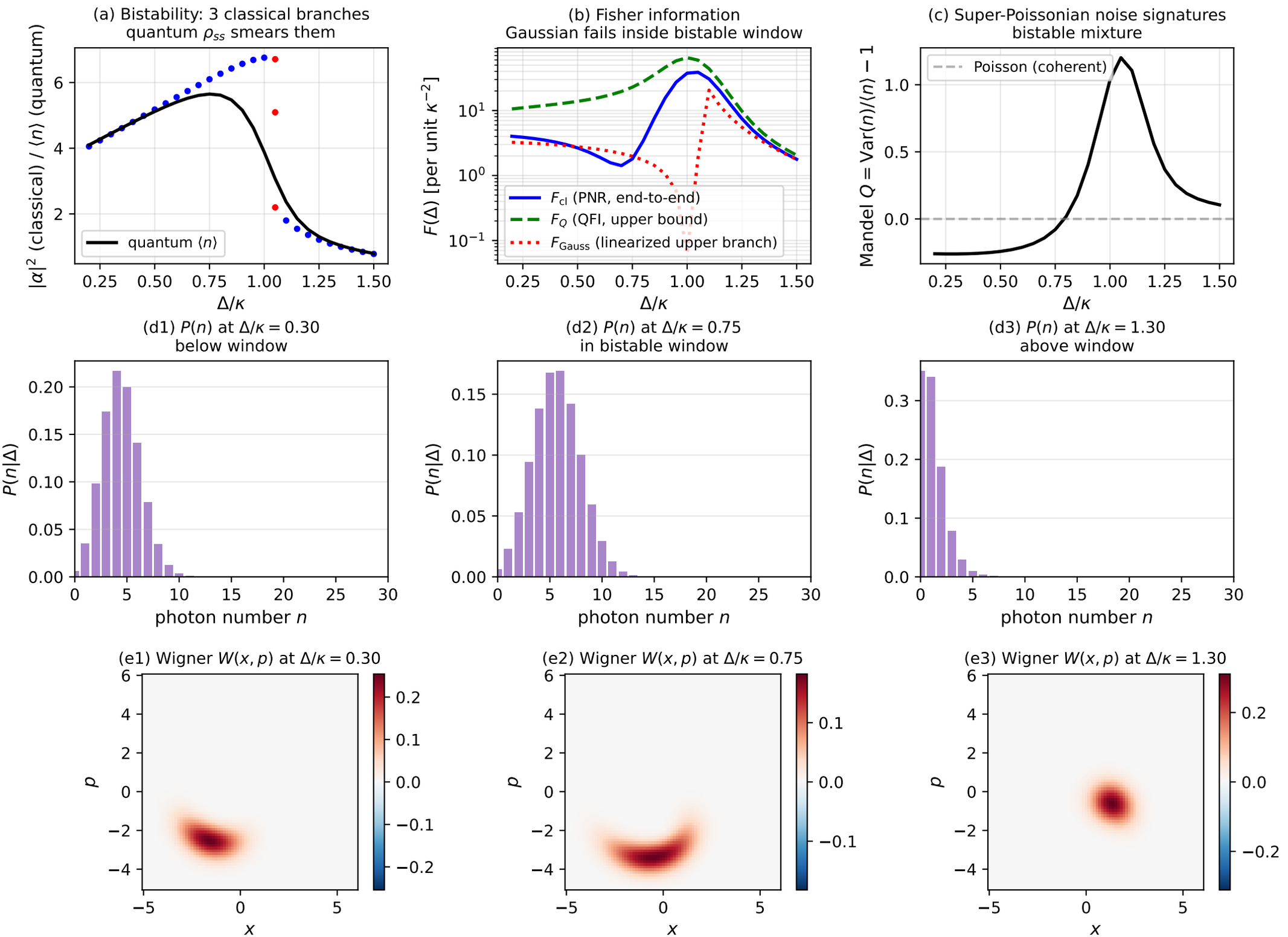
End-to-end Keldysh Fisher information without linearization
(bistable driven Kerr cavity -- Gaussian approximation provably fails)
(a) Bistability: 3 classical branches
quantum ρ_{ss} smears them
$|\alpha|^2$ (classical) / $\langle n \rangle$ (quantum)
quantum $\langle n \rangle$
Δ/κ
(b) Fisher information
Gaussian fails inside bistable window
$F(\Delta)$ [per unit κ^{-2}]
$F_{\rm cl}$ (PNR, end-to-end)
F_Q (QFI, upper bound)
$F_{\rm Gauss}$ (linearized upper branch)
Δ/κ
(c) Super-Poissonian noise signatures
bistable mixture
Mandel $Q = {\rm Var}(n)/\langle n \rangle - 1$
Poisson (coherent)
Δ/κ
(d1) $P(n)$ at $\Delta/\kappa = 0.30$
below window
$P(n|\Delta)$
photon number n
(d2) $P(n)$ at $\Delta/\kappa = 0.75$
in bistable window
$P(n|\Delta)$
photon number n
(d3) $P(n)$ at $\Delta/\kappa = 1.30$
above window
$P(n|\Delta)$
photon number n
(e1) Wigner $W(x,p)$ at $\Delta/\kappa = 0.30$
(e2) Wigner $W(x,p)$ at $\Delta/\kappa = 0.75$
(e3) Wigner $W(x,p)$ at $\Delta/\kappa = 1.30$
x
p